

Lily Oglesby

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Purpose

Graduate student in business research with a background in data science and astrophysics. Interested in using data to understand and solve real-world problems and support better decisions across industries. Experience building Python and SQL workflows, training machine-learning models, and analyzing scientific, public health, and business data.

Skills

- **Programming and Data Science:** Python, SQL, JavaScript, HTML, CSS, Pandas, NumPy, SciPy, Scikit-Learn, TensorFlow/Keras, PyMC, Matplotlib, Seaborn, Jupyter
- **Databases and Development:** PostgreSQL, MongoDB, Git, GitHub, Linux, Docker, VS Code, LaTeX
- **Analytical Methods:** Machine Learning, Statistical Modeling, Regression, Random Forests, Neural Networks, CNNs, Anomaly Detection, Time-Series Analysis, Bayesian Inference, Causal Inference, Cross-Validation, Feature Engineering, Model Evaluation, Data Visualization
- **Communication and Collaboration:** Clear written and verbal communication; translates complex technical results for technical and nontechnical audiences; manages multiple priorities and changing requirements; works well independently and as a team member; experienced in teaching and mentoring learners of all ages

Education

- **University of Southern California | Marshall School of Business** Los Angeles, CA
Master of Science in Business Research Aug 2025 – Dec 2026
 - **Upcoming Fall 2026 coursework:** Business Analytics, AI Modeling for Decision-Making, Regression Analysis, Spreadsheet Modeling, Data Analytics; Tableau and SAS coursework in progress.
- **University of California, Berkeley** Berkeley, CA
Bachelor of Arts: Double Major in Data Science and Astrophysics Aug 2021 – May 2025

Selected Data Science and Engineering Projects

- **Interactive Portfolio Game** 2026
JavaScript, HTML/CSS, Canvas, GSAP, Howler.js, Tiled
 - Built and debugged a responsive portfolio website presented as a browser game, with an explorable map, collision detection, turn-based battles, mobile controls, animation, audio, asset preloading, and Netlify deployment.
- **Haqdarshak Social Protection Access Consulting Project** 2025
USC Graduate Research, Program Design, Stakeholder Analysis
 - Worked with a faculty advisor to explore ways Haqdarshak field workers could connect more eligible individuals with government benefit and subsidy programs; reviewed operational challenges and developed preliminary ideas for improving outreach and service delivery.
- **Scientific Data Analytics Portfolio** 2024
Python, Pandas, NumPy, Scikit-Learn, PyMC, Data APIs, Astropy
 - Built reproducible workflows using satellite observations, stellar time-series data, telescope images, and astronomical spectra; applied database queries, feature engineering, regression, cross-validation, Bayesian estimation, high-dimensional preprocessing, and model evaluation.
- **Particulate Matter Health Impact and Forecasting** 2024
Python, Pandas, Scikit-Learn, Causal Inference
 - Analyzed CDC and NOAA data to study PM2.5 exposure and asthma outcomes and predict pollution from weather; compared generalized linear and random forest models using leakage-resistant temporal validation.
- **PostgreSQL vs. MongoDB Yelp Benchmark** 2024
PostgreSQL, MongoDB, Python, Jupyter, Docker
 - Designed equivalent workloads across relational and document databases and compared data models, SQL queries, aggregation pipelines, indexes, query plans, scan volume, and runtime performance.

Research Experience

- **Space Sciences Laboratory, UC Berkeley** Berkeley, CA
Undergraduate Researcher - Data Analysis Aug 2021 – May 2025
 - Built reusable Python pipelines for three years of satellite data, applying time-series preprocessing, statistical filtering, and visualization to identify event-related patterns and support a [peer-reviewed publication](#) in the *Journal of Geophysical Research*.
- **Institute for Astronomy, University of Hawai'i at Manoa** Honolulu, HI
Machine Learning Research Intern Jun 2024 – Aug 2024
 - Applied anomaly detection to approximately 800,000 astronomical images, trained a convolutional neural network for multiclass classification, and used GPU parallelization to accelerate preprocessing and model execution. [Report available online.](#)

Teaching and Mentoring Experience

- **USC, UC Berkeley, and University of Hawai'i** Los Angeles, Berkeley, and Honolulu
Instructor, Teaching Assistant, Reader, Grader, and Research Mentor Aug 2022 – Jul 2026
 - Taught and mentored high school, undergraduate, and graduate students through USC, UC Berkeley, and the Mauna Kea Scholars Program; led workshops and office hours on responsible AI, Python, data analysis, research design, graduate applications, and technical presentations, adapting instruction to students' backgrounds and experience levels.